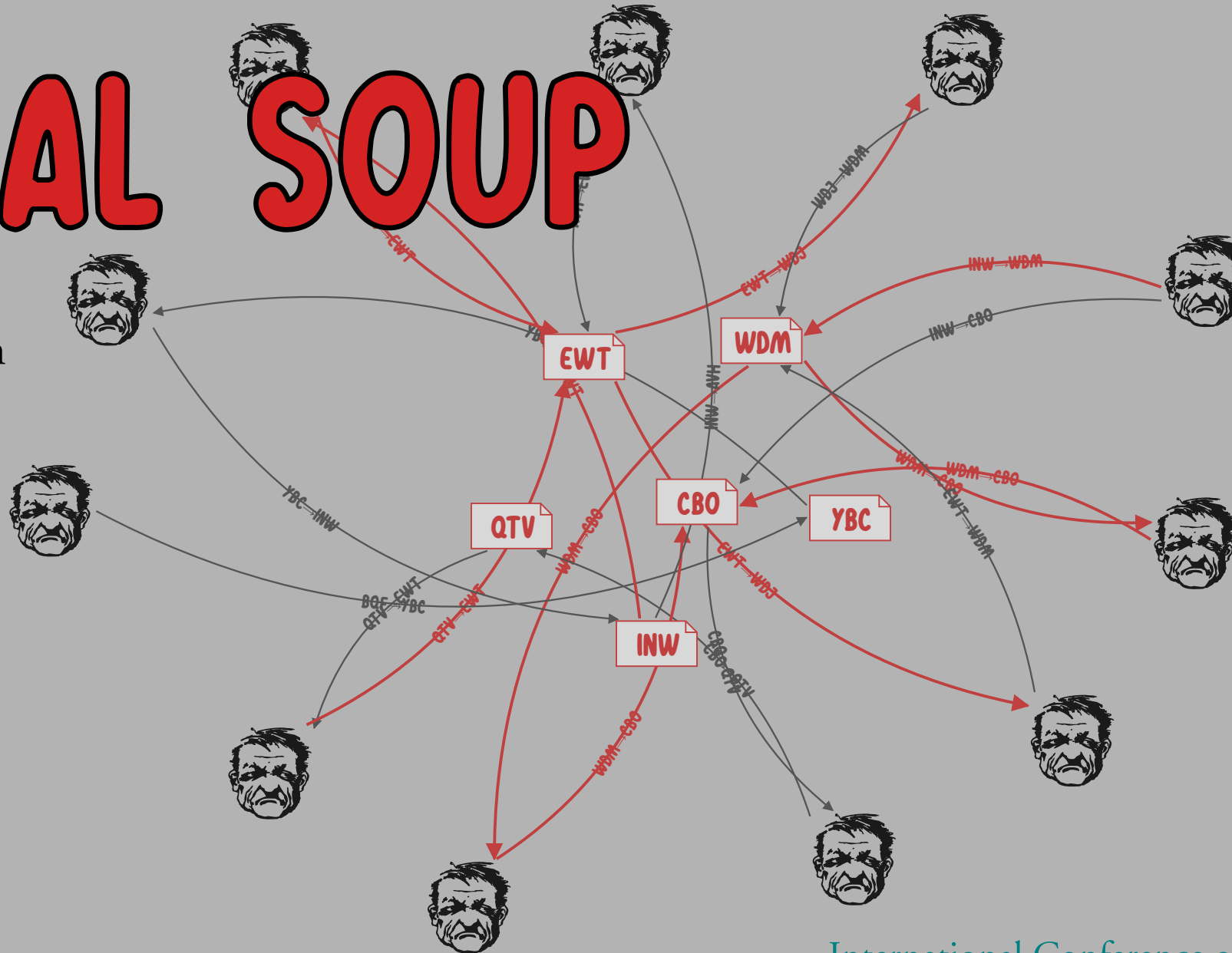


SIGNAL SOUP

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DI TORINO



→ SHARED SF → UNIQUE SF

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SIGNAL SOUP

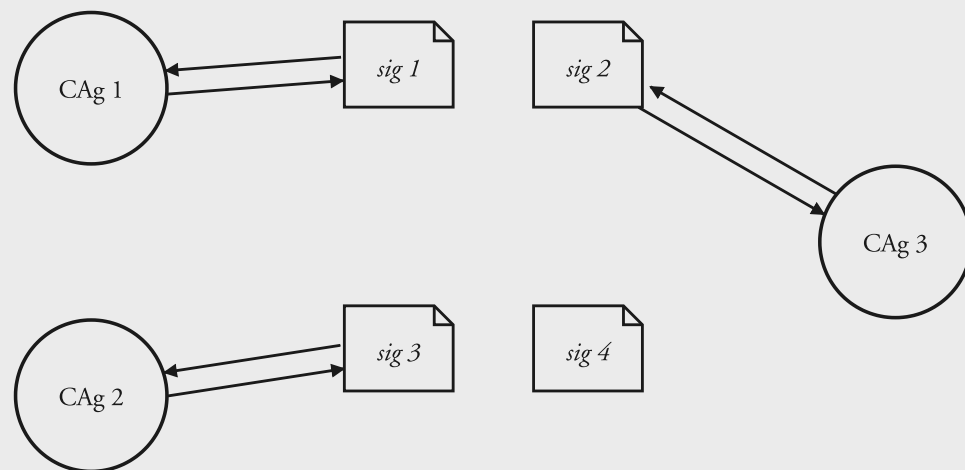
We introduce Signal Soup, a simple model emphasizing "loose" code alignment —
a Gedankenexperiment formalized for in silico simulation.

LEXICAL COORDINATION MODELS ASSUME...

- **Arbitrariness** — no motivated link between the two planes (*Saussure 1916*)
- **Direct feedback** — success is evaluated through immediate sender/receiver exchange
- **Dyadic relationship** — communication modelled as pairs, even in larger groups
- **Optimal alignment** — ideal communication = complete code sharing among all participants

01

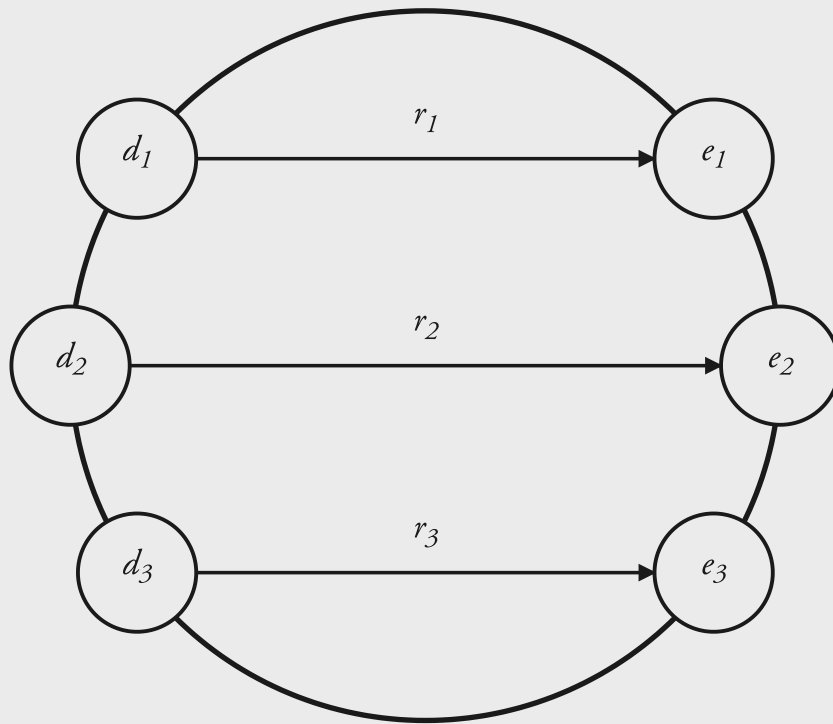
SIGNAL SOUP — THE MODEL



CENV + CAGS

COMMUNICATIVE ENVIRONMENT

- **Communication** = interaction among CAgS mediated by a shared communicative environment (*CEnv*)
- **Signal** — 3-letter word, lowercase alphabet —
 $26^3 = 17,576$ possible signals
- Agents process and *replace* signals
- No direct agent-to-agent exchange

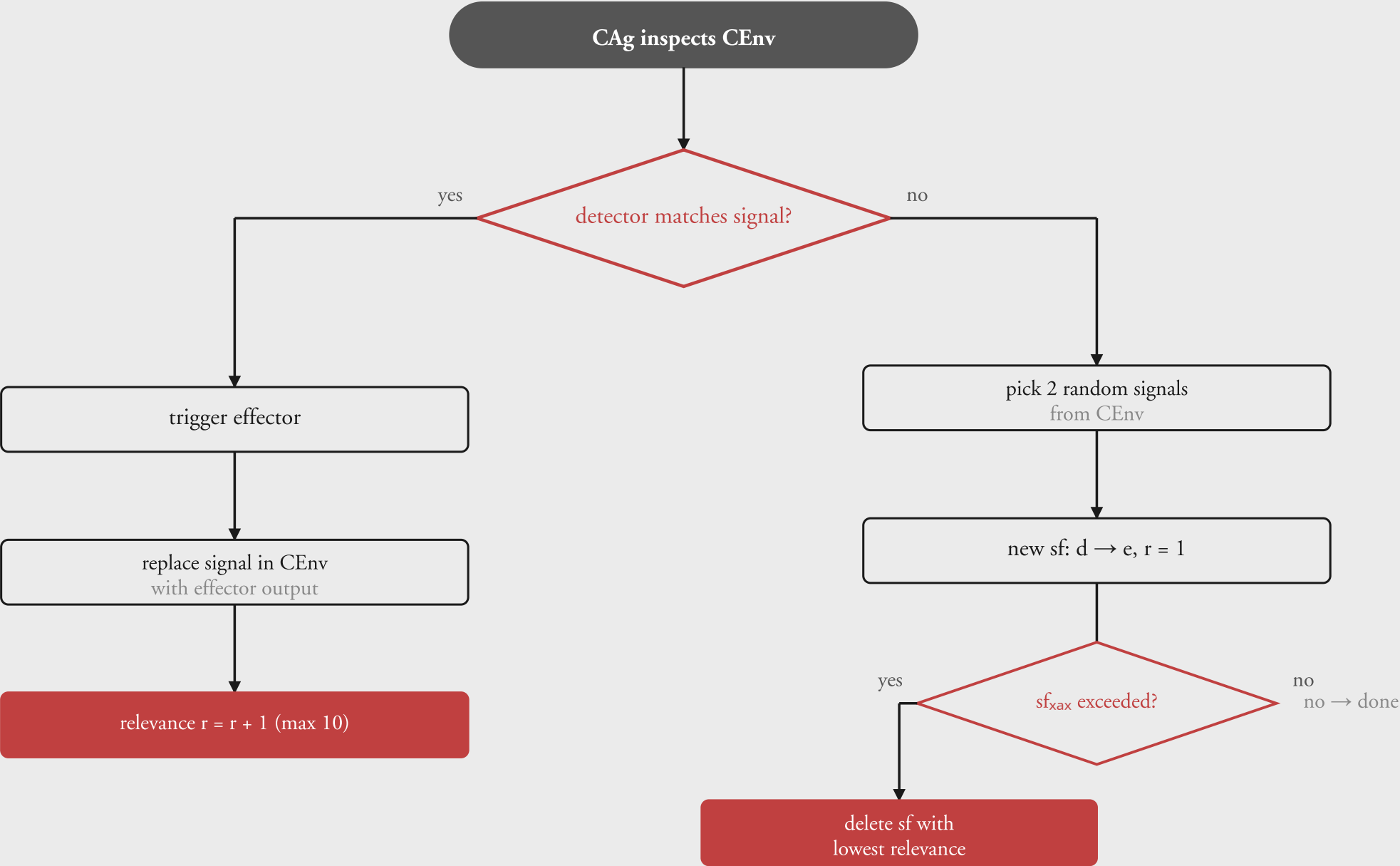


CAG STRUCTURE

COMMUNICATIVE AGENT

- **Detector** $\rightarrow$ **Effector** — 3-letter signal types
(Holland 1995)
- **Sign function** sf — mapping $d \rightarrow e$
- **Relevance** r — score per sf , capped at $r_{\max} = 10$
- **Memory** — $sf_{\max} = 10$ sign functions per agent

THE ALGORITHM



02

THE BAND

THE BAND

- **Band** — n CAgS sharing a single CEnv | CEnv size = n
- **Initialisation** — each CAg gets 1 sf (random $d \rightarrow e$), with $r = 0$ (*easily replaced*)
- **CEnv seed** — one effector signal per agent; agents start by emitting their only output
- **Asynchronous** — at each turn, one randomly selected CAg inspects CEnv and acts

03

WHAT EMERGES

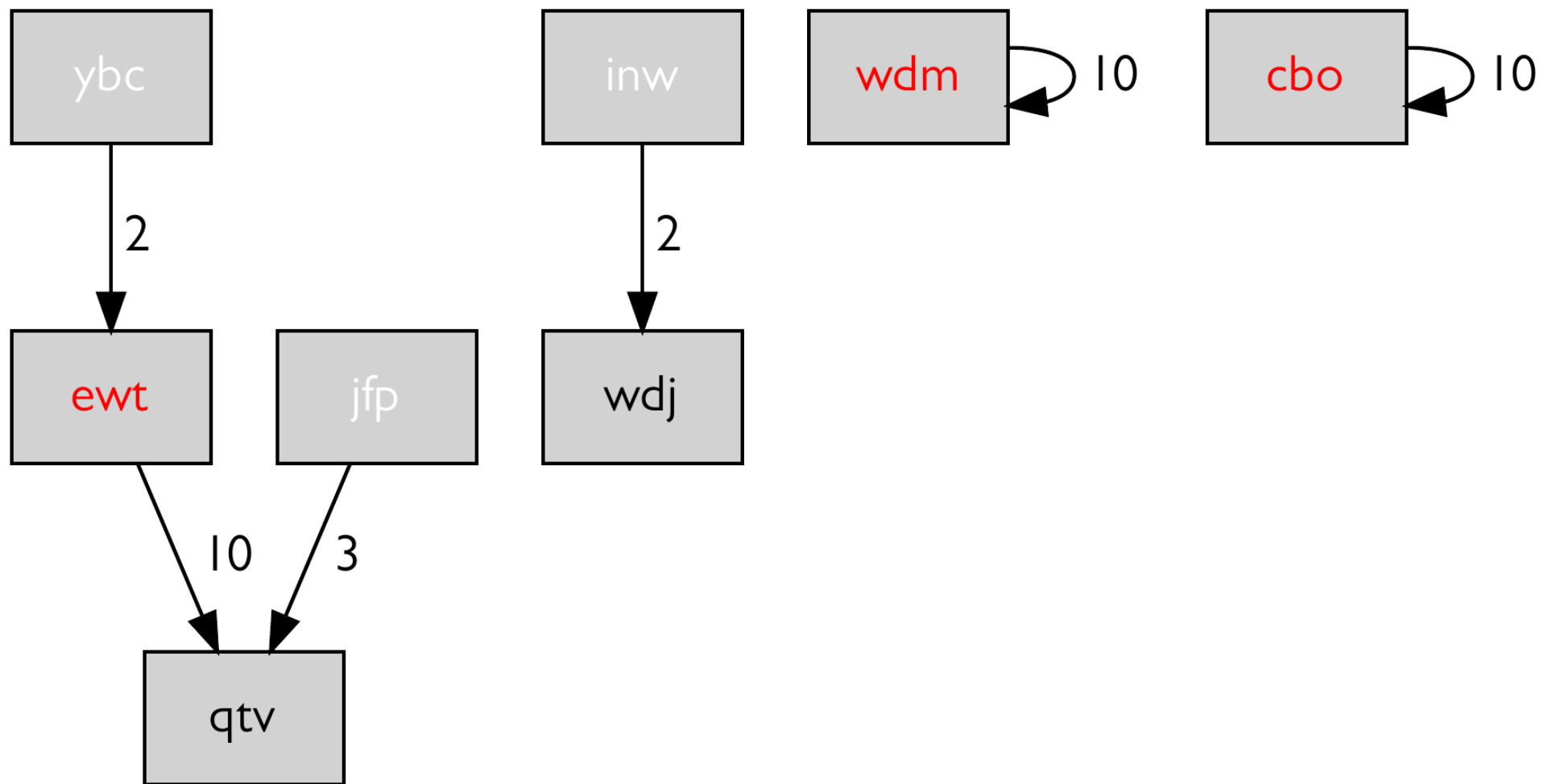


FIG. 2 CAg topology after 2,000 turns — loops, chains, synonymy emerge spontaneously

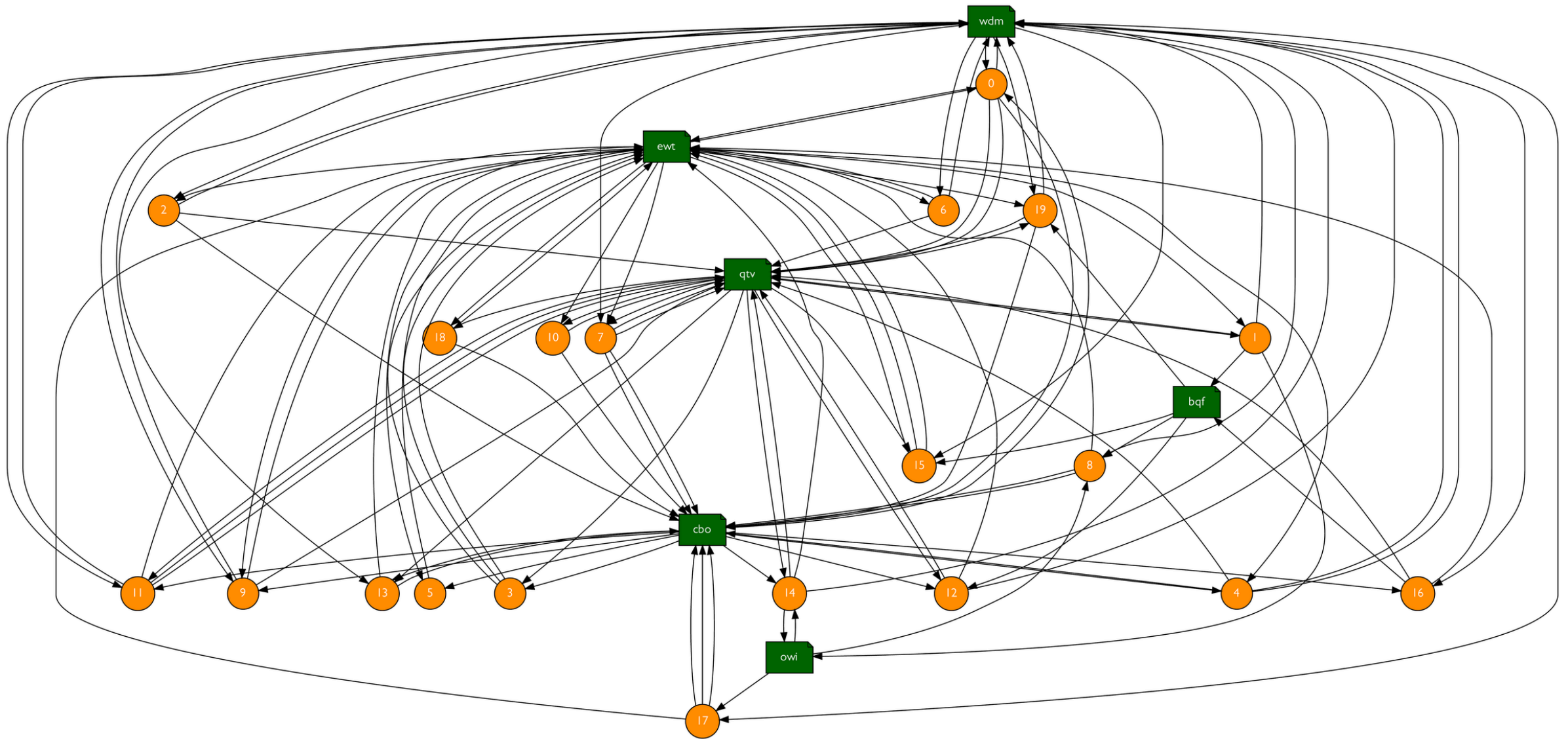


FIG. 3 Band-CEnv matching relations — 20 agents, 2,000 turns

DEFINITION

Type-token ratio = *distinct signal types* / *total signal occurrences* in CEnv

ratio = 1 $\rightarrow$ all signals distinct · ratio = 0 $\rightarrow$ high redundancy

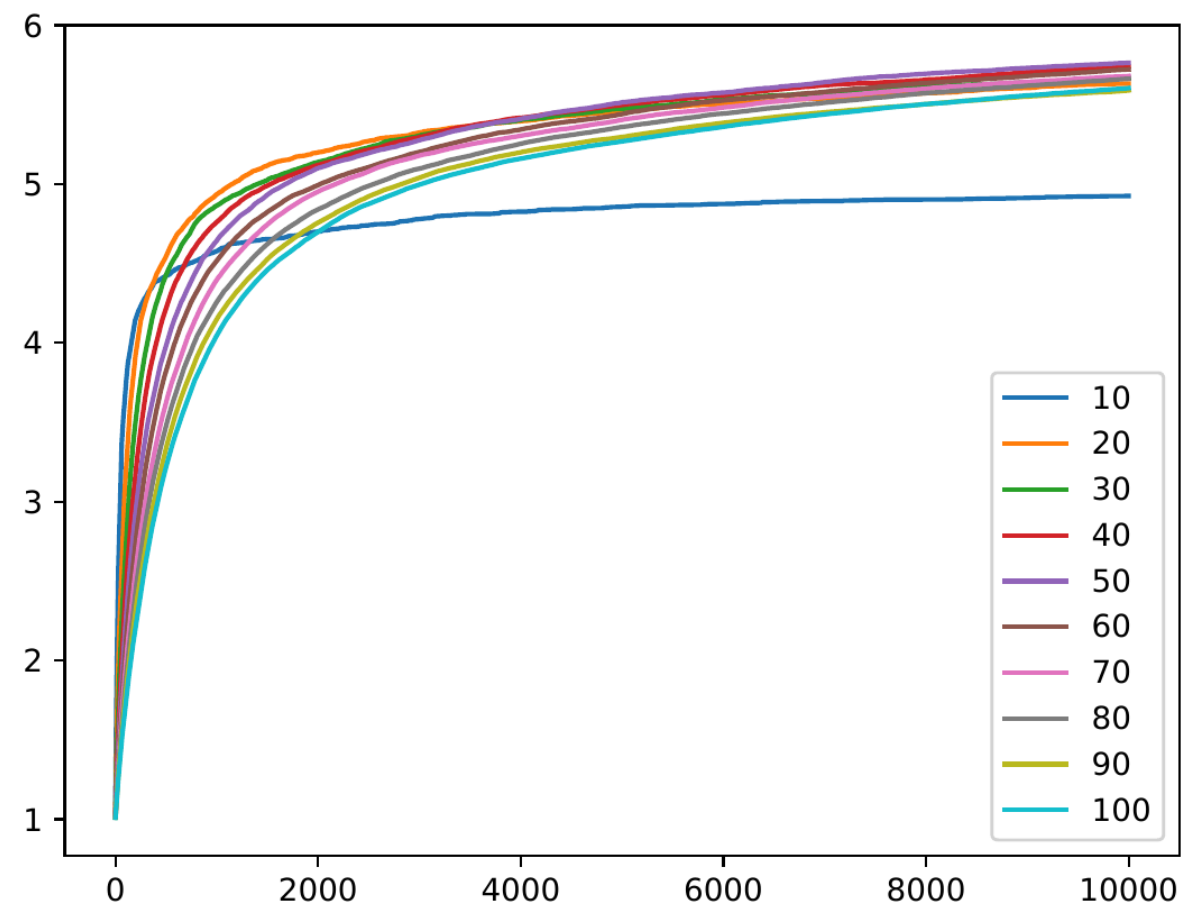
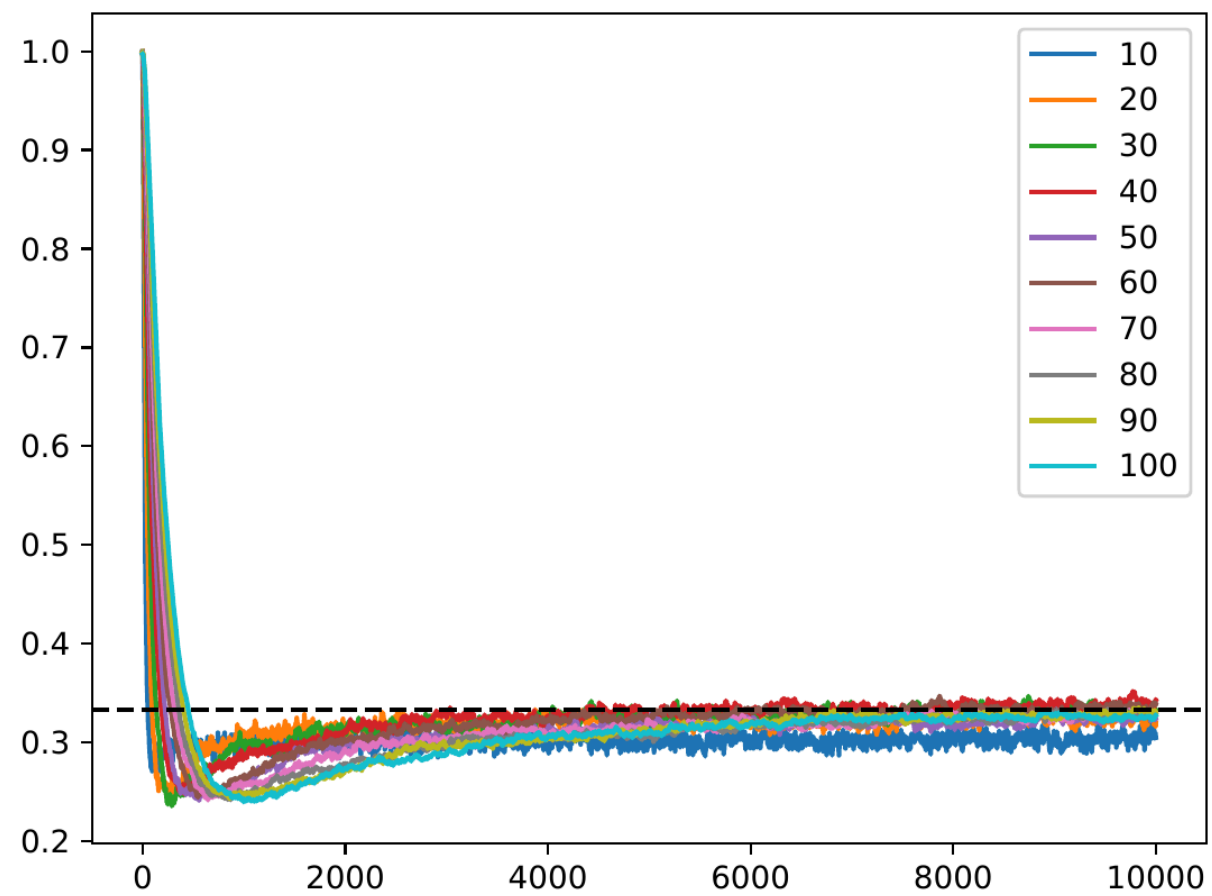


FIG. 4 Type-token ratio $\rightarrow 1/3$ · avg sf/agent $\rightarrow \sim 5.6$ · 10 band sizes, 100 runs each, 10,000 turns

PARTIAL ALIGNMENT

- After 2,000 turns: ~**26%** of sign functions shared by 2+ agents
- Some shared by **up to 7** agents — without any centralised coordination
- Majority of sf remain **idiosyncratic** — each agent keeps its own repertoire
- Eco: not a *dictionary* (shared code) but an *encyclopedia* (distributed competence)

04

DISCUSSION

THREE ASSUMPTIONS — REVISITED

- **Direct feedback** → *none* — agents act on CEnv, never receive success/failure signals
- **Dyadic** → pairs, dialogical vs *n agents, polylogical*
- **Optimal alignment** → complete code sharing vs *sharedness* — *partial overlap*

STIGMERGY

- Agents coordinate *indirectly* — by modifying and responding to CEnv
- No direct agent-to-agent exchange — the environment is the medium *and* the message
- Priming through repeated exposure: signals that appear often get reinforced (*Pickering & Garrod 2004*)
- Local interactions → global patterns (*Theraulaz & Bonabeau 1999*)

THE BAND AS SOCIAL UNIT

- Not pairs but a *small, tightly integrated population* — a social unit (*Fitch 2010*)
- Detection and action are **mandatory** — every agent acts on every turn
- This embodies *Mitteilungsbedürfnis* — the compulsion to communicate (*Fitch 2010*)
- Akin to enforced Gricean cooperation — not chosen, but structural (*Grice 1989*)

CONNECTIVITY, NOT ALIGNMENT

- Graph connectivity + circuit formation (*Steels 1997*) recalls **autocatalytic sets** (*Kauffman 2000, 2019*)
- Looping sf yield *copier* behaviour (*Fontana 1991*) — ~ 5.6 sf/agent is the threshold

ENCYCLOPEDIA, NOT DICTIONARY

- Not a shared code but an idiosyncratic, overlapping repertoire (*Eco 1986*)
- Distributed competence — sufficient for coordination, without convergence

1/3

type-token ratio

*three tokens per signal type in CEnv
all band sizes · all runs*

5.6

avg sign functions / agent

*well below $sf_{max} = 10$
sufficient for coordination*

TWO EMERGENT EQUILIBRIA — WHY?

- **Reinforcement dynamics** — frequent signals match more, get reproduced — positive feedback
- **Emergent repertoire size** — ~ 5.6 sf is sufficient for coordination; rare signals are replaced
- **Stigmergic coordination** — agent A produces $xyz \rightarrow$ raises probability agent B learns xyz
- Result: $tp:tk \rightarrow 1/3$ · sf/agent $\rightarrow \sim 5.6$ · robust across $n = 10-100$

CONCLUSIONS FROM A SIGNAL SOUP

Stable communication emerges *without* direct feedback, dyadic exchange, or complete code sharing.

Stigmergy + partial alignment = sufficient for coordination (connectivity).

Two emergent constants — $tp:tk \rightarrow 1/3 \cdot sf/agent \rightarrow 5.6$ — mark the connectivity threshold.

Sharedness, not consensus, is the condition.

GRAZIE

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github.com/vanderaalle/signal-soup

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